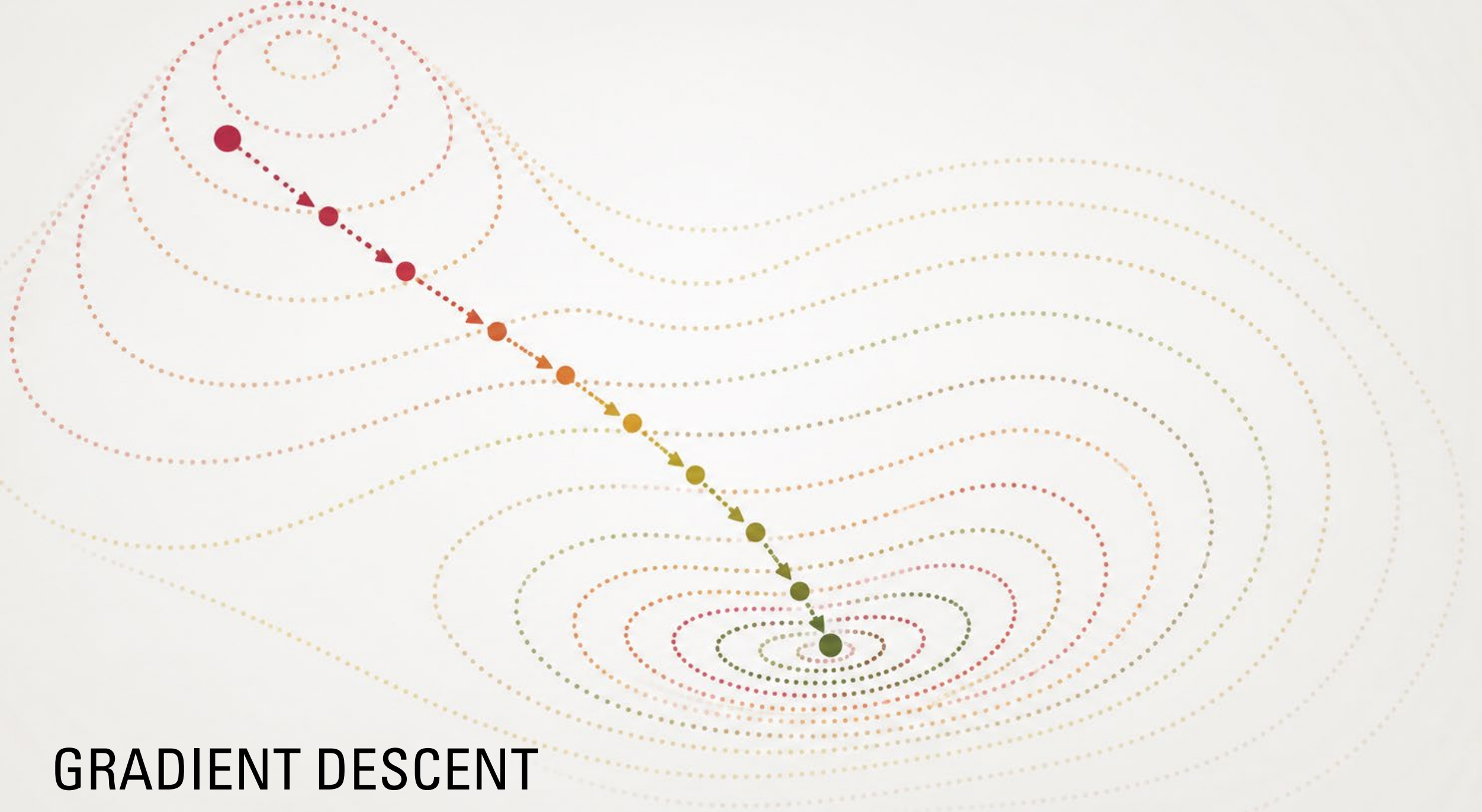




GRADIENT DESCENT

OPTIMIZATION IN MACHINE LEARNING

- Many (all?) machine learning problems can be posed as a minimization or maximization problem
- In some cases, a closed-form solution exists
 - E.g., linear regression
- Most of the time, a closed-form (analytic) solution doesn't exist
 - E.g., logistic regression
- We can solve these problems using optimization theory

UNCONSTRAINED OPTIMIZATION

- An *unconstrained optimization problem* has the form

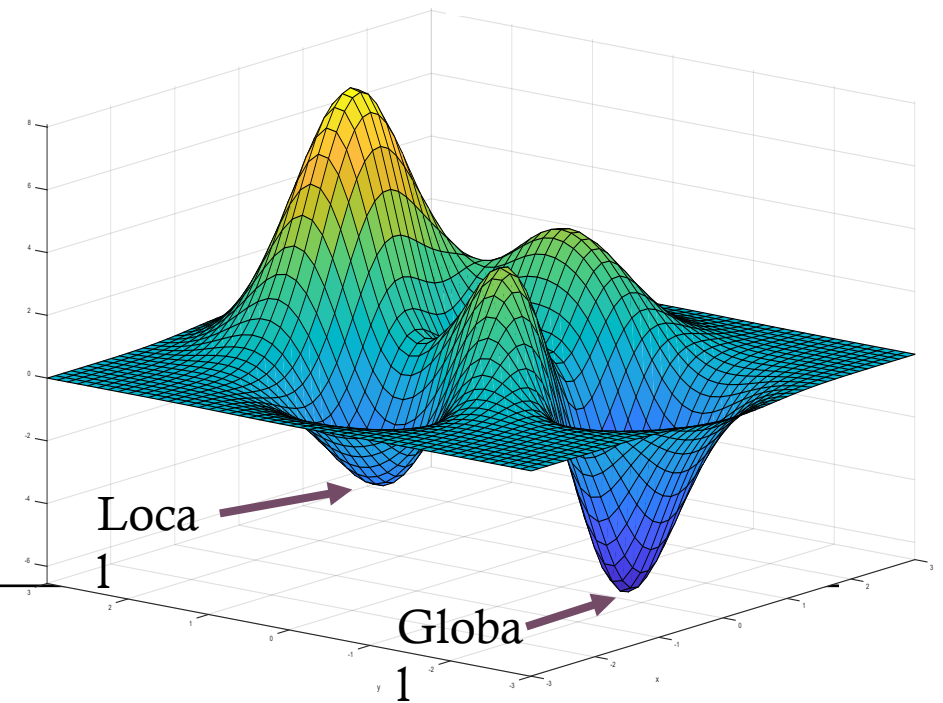
$$\min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$$

where $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is called the *objective function*.

- What about maximization?
- A point $\mathbf{x}^* \in \mathbb{R}^d$ is called a *local minimizer* if $\exists r > 0$ such that
 - $f(\mathbf{x}^*) \leq f(\mathbf{x}) \forall \mathbf{x}$ satisfying $\|\mathbf{x} - \mathbf{x}^*\| < r$
- $\mathbf{x}^*$ is called a *global minimizer* if
 - $f(\mathbf{x}^*) \leq f(\mathbf{x}) \forall \mathbf{x} \in \mathbb{R}^d$
- Is a global minimizer also a local minimizer?

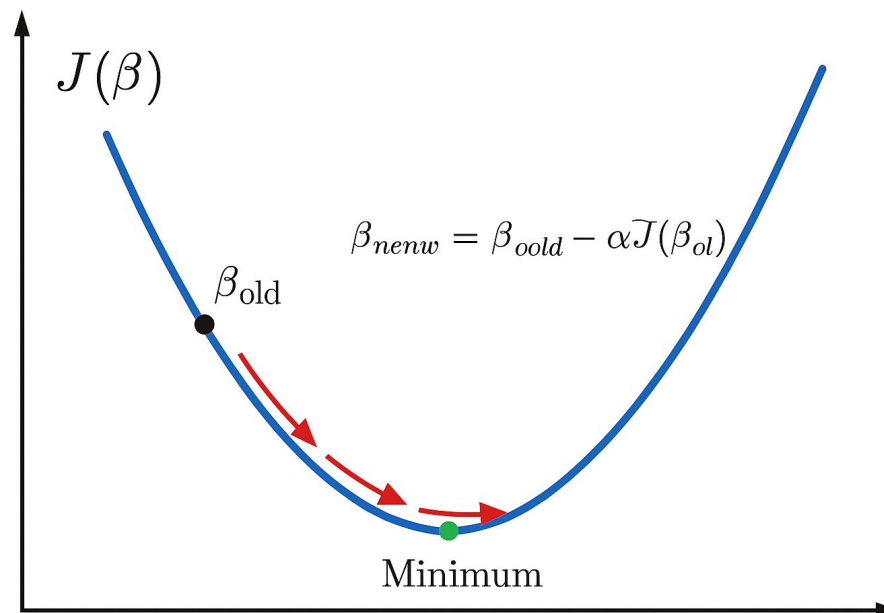
$\exists$ = “there exists”

$\forall$ = “for all”



GRADIENT DESCENT (HIGH LEVEL)

- Numerical optimization algorithm that tweaks parameters iteratively until the minimum is found
 - “Descend” the gradient of the cost function in the steepest direction until the minimum is reached
- What is the gradient?

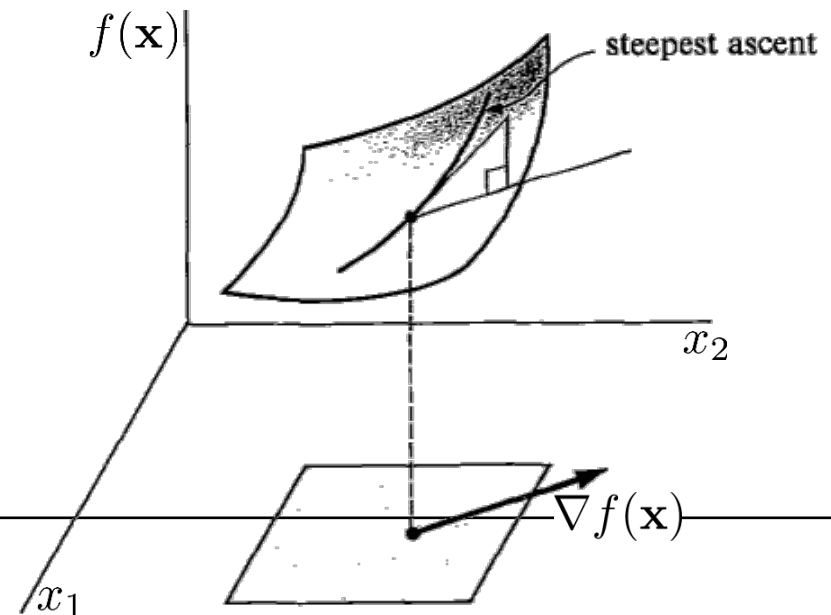


GRADIENT

- Given a function $f: \mathbb{R}^d \rightarrow \mathbb{R}$, the *gradient* of f at $\mathbf{x} = [x_1 \dots x_d]^T \in \mathbb{R}^d$ is defined by

$$\nabla f(\mathbf{x}) := \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_d} \end{bmatrix}$$

- If the gradient exists for all $\mathbf{x} \in \mathbb{R}^d$, we say f is *differentiable*
- The gradient gives the direction of *steepest ascent*



FIRST ORDER NECESSARY CONDITION

- If f is differentiable and $\mathbf{x}^*$ is a local minimizer of f , then
$$\nabla f(\mathbf{x}^*) = \mathbf{0}.$$
 - Note that this condition is necessary but not sufficient for $\mathbf{x}^*$ to be a local minimizer
 - Why?
 - If $\nabla f(\mathbf{x}) = 0$ for some $\mathbf{x}$, then $\mathbf{x}$ is said to be a critical point or a stationary point of f
-

HESSIAN

- The Hessian of f at $\mathbf{x}$ is the $d \times d$ matrix

$$\nabla^2 f(\mathbf{x}) := \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_d \partial x_1} & \cdots & \frac{\partial^2 f(\mathbf{x})}{\partial x_d^2} \end{bmatrix}$$

- We say that f is twice differentiable if $\nabla^2 f(\mathbf{x})$ exists $\forall \mathbf{x} \in \mathbb{R}^d$
- We say f is twice continuously differentiable if it is twice differentiable and all of the second derivatives are continuous functions of $\mathbf{x}$
- If f is twice continuously differentiable, then $\nabla^2 f(\mathbf{x})$ is a symmetric matrix $\forall \mathbf{x}$, i.e.,

$$\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 f(\mathbf{x})}{\partial x_j \partial x_i}$$

$$\forall \mathbf{x} \in \mathbb{R}^d, \forall i, j = 1, \dots, d$$



POSITIVE (SEMI-)DEFINITE MATRICES

- Let A be a $d \times d$ matrix. We say that A is *positive definite* (PD) if $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$. We say that A is *positive semi-definite* (PSD) if $\mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^d$.
 - PD and PSD arise frequently in ML, for example
 - Gram matrices (i.e., XX^T)
 - Kernel matrices
 - Covariance matrices
 - Hessian matrices (sometimes)
 - PD/PSD matrices are not necessarily symmetric, e.g.
$$\begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x & y \end{bmatrix} = x^2 + y^2$$
 - However, we will only consider PD/PSD matrices that are also symmetric
-

SECOND ORDER NECESSARY CONDITION

- If f is twice continuously differentiable and $\mathbf{x}^*$ is a local min, then $\nabla^2 f(\mathbf{x}^*)$ is positive semi-definite, i.e.,

$$\mathbf{z}^T \nabla^2 f(\mathbf{x}^*) \mathbf{z} \geq 0, \quad \forall \mathbf{z} \in \mathbb{R}^d$$

- This generalizes the result from single-variable calculus that the second derivative is nonnegative at a local min
- Note: the converse is *not* true! (If PSD, then local min.. not guaranteed!)
 - However, if not PSD, then not min!



LEADING PRINCIPAL MINORS

- Assume A is a real symmetric $n \times n$ matrix.
 - The k th leading principal minor is $\Delta_k = \det(A_{1:k, 1:k})$, the determinant of the upper-left $k \times k$ block.
 - **PD (Sylvester's criterion):** A is positive definite if and only if $\Delta_1 > 0, \Delta_2 > 0, \dots, \Delta_n > 0$.
 - **PSD:** A is positive semidefinite if and only if every principal minor is nonnegative.
 - A principal minor uses the same subset of row and column indices, not necessarily the first k .
 - **Caution:** $\Delta_k \geq 0$ (leading) for every k is insufficient for PSD. For example, $\text{diag}(0, -1)$ has leading minors 0 and 0, but is not PSD.
-



WORKED EXAMPLE: LEADING MINORS

- Consider the real symmetric 2×2 matrix

$$A(t) = \begin{bmatrix} 2 & 1 \\ 1 & t \end{bmatrix} \quad (t \text{ is real})$$

- **First leading minor:** $\Delta_1 = \det([2]) = 2 > 0$.
- **Second leading minor:** use $ad - bc$ for the determinant.

$$\Delta_2 = \det(A(t)) = \begin{vmatrix} 2 & 1 \\ 1 & t \end{vmatrix} = (2)(t) - (1)(1) = 2t - 1$$

- **PD conclusion:** both leading minors must be positive.
Since $\Delta_1 = 2 > 0$, we need $2t - 1 > 0$, giving $t > 1/2$.
 - **For example, $t = 1$:** $\det(A(1)) = 2(1) - 1 = 1 > 0$,
so $A(1)$ is PD.
-



EXAMPLE: THE PSD CHECK

- For $A(t)$, the three principal minors are $a_{11} = 2$, $a_{22} = t$, and $\det(A(t)) = 2t - 1$. PSD requires all three to be ≥ 0 .
- At the boundary $t = 1/2$:

$$A(1/2) = \begin{bmatrix} 2 & 1 \\ 1 & 1/2 \end{bmatrix}$$

$$\det(A(1/2)) = \begin{vmatrix} 2 & 1 \\ 1 & 1/2 \end{vmatrix} = (2)(1/2) - (1)(1) = 1 - 1 = 0$$

- **PSD check:** the principal minors are 2, $1/2$, and 0. All are nonnegative, so $A(1/2)$ is PSD. The zero determinant means it is not PD.
 - **For $t < 1/2$:** $\det(A(t)) = 2t - 1 < 0$, so $A(t)$ is neither PD nor PSD.
 - **Summary:** $A(t)$ is PD for $t > 1/2$ and PSD for $t \geq 1/2$.
-

GROUP EXERCISE

Notation:

$$\boldsymbol{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Consider the function

$$f(x, y) = x^2 + 4xy - y^2 - 8x - 6y + 10$$

1. Determine $\nabla f(x, y)$
 2. Determine $\nabla^2 f(x, y)$
 3. Determine a critical point $\boldsymbol{x}^*$
 4. Is $\boldsymbol{x}^*$ a local min, a local max, or neither?
-

CONVEXITY

- We say that f is *convex* if

$$f(t\mathbf{x} + (1 - t)\mathbf{y}) \leq tf(\mathbf{x}) + (1 - t)f(\mathbf{y})$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and $t \in [0,1]$.

- We say f is *strictly convex* if

$$f(t\mathbf{x} + (1 - t)\mathbf{y}) < tf(\mathbf{x}) + (1 - t)f(\mathbf{y})$$

$\forall \mathbf{x} \neq \mathbf{y}$ and $t \in (0,1)$.

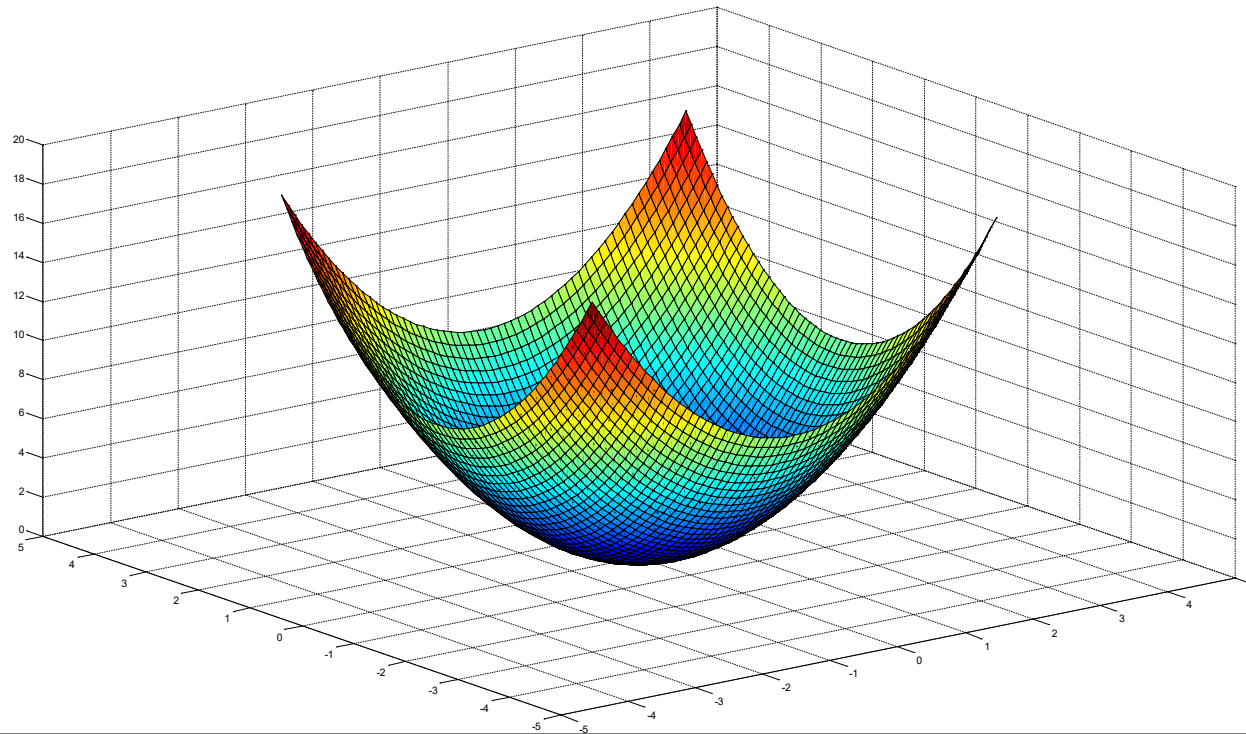
CONVEX FUNCTIONS ARE NICE

Properties of convex functions

1. If f is convex, then every local min is a global min
2. If f is strictly convex, then f has at most one global min

CONVEXITY EXAMPLE

$$f(x, y) = x^2 + y^2$$



HESSIAN, CONVEXITY, AND CRITICAL POINTS

Assume f is twice continuously differentiable on an open convex domain D .

1. A PSD Hessian throughout D makes f convex.

For every x in D and direction v : $v^T \nabla^2 f(x) v \geq 0$.

2. A convex function lies above each tangent plane.

$f(y) \geq f(x^*) + \nabla f(x^*)^T (y - x^*)$ for every y in D .

3. At a critical point, $\nabla f(x^*) = 0$, so $f(y) \geq f(x^*)$.

Therefore, x^* is a global minimizer of f on D .

Caution: PSD only at x^* is inconclusive. For $f(x) = x^3$, $f'(0) = f''(0) = 0$, but 0 is not a local minimum.

GROUP EXERCISE

1. Give an example of a function f that is
 - convex but not strictly convex
 - convex and has more than one global minimum
 - strictly convex, but has no global minimum
 2. Is the sum of convex functions necessarily convex? Prove or provide a counter example
 3. Is the product of convex functions necessarily convex? Prove or provide a counter example
-

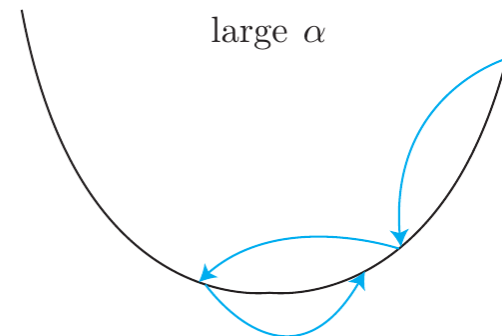
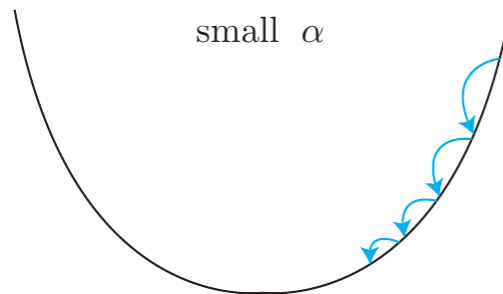
GRADIENT DESCENT (GD)

- Consider minimizing the generic objective function $J(\theta)$
- What is a geometric interpretation of $\nabla J(\theta)$?
 - Direction of steepest ascent
 - Thus iterative approaches take steps in opposite direction, i.e., the direction of steepest descent
- Initial guess θ_0
- Learning rate, α
- For $t = 1, \dots, \text{max_iter}$

$$\theta_t \leftarrow \theta_{t-1} - \alpha \nabla J(\theta_{t-1})$$

If convergence condition satisfied, exit

End

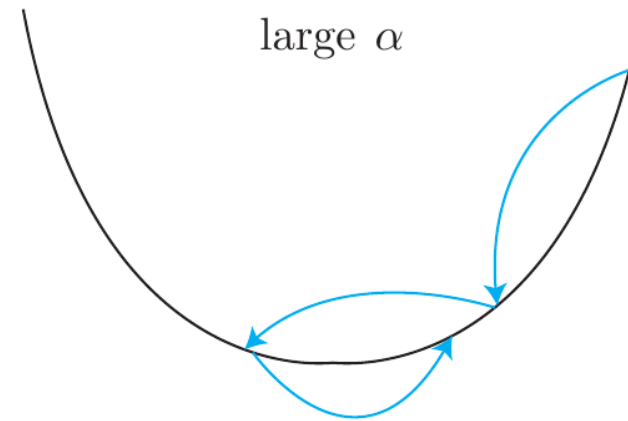
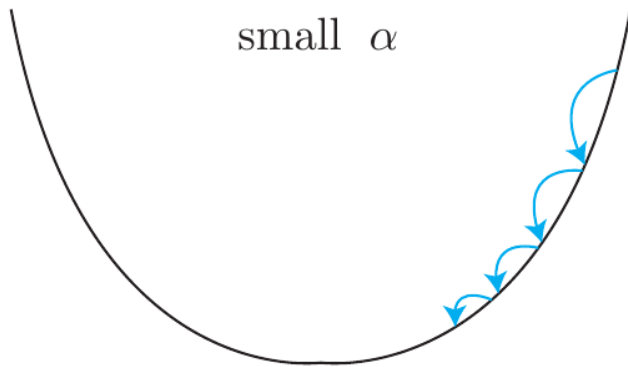


LEARNING RATE

- If the learning rate is too high, the gradient descent algorithm will bounce around the minimum and never converge
 - (See Figure 4-5 in HOML)
- If the learning rate is too low, it could take a very long time to find the minimum
 - (See Figure 4-4 in HOML)
- It is common to start with a small learning rate (~ 0.01) and adjust based on the performance of the model

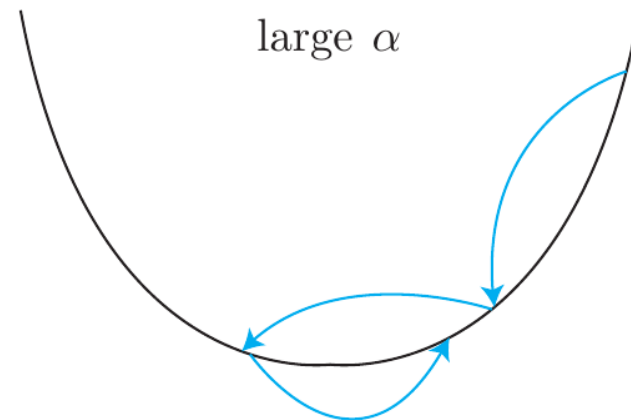
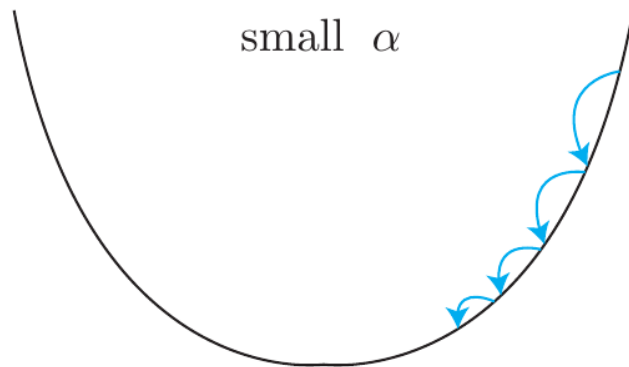
LEARNING RATE AS A FUNCTION

- In practice, typically need α to decrease as a function of t
 - Example: $\alpha_t = \frac{1}{t}$



GROUP EXERCISE

1. What would be a good convergence condition (also known as a stopping criterion) for terminating GD?
2. What are some potential problems with setting $\alpha_t = \frac{1}{t}$?
3. How could you choose α_t to counter these problems?
4. Does the initialization θ_0 affect the result?



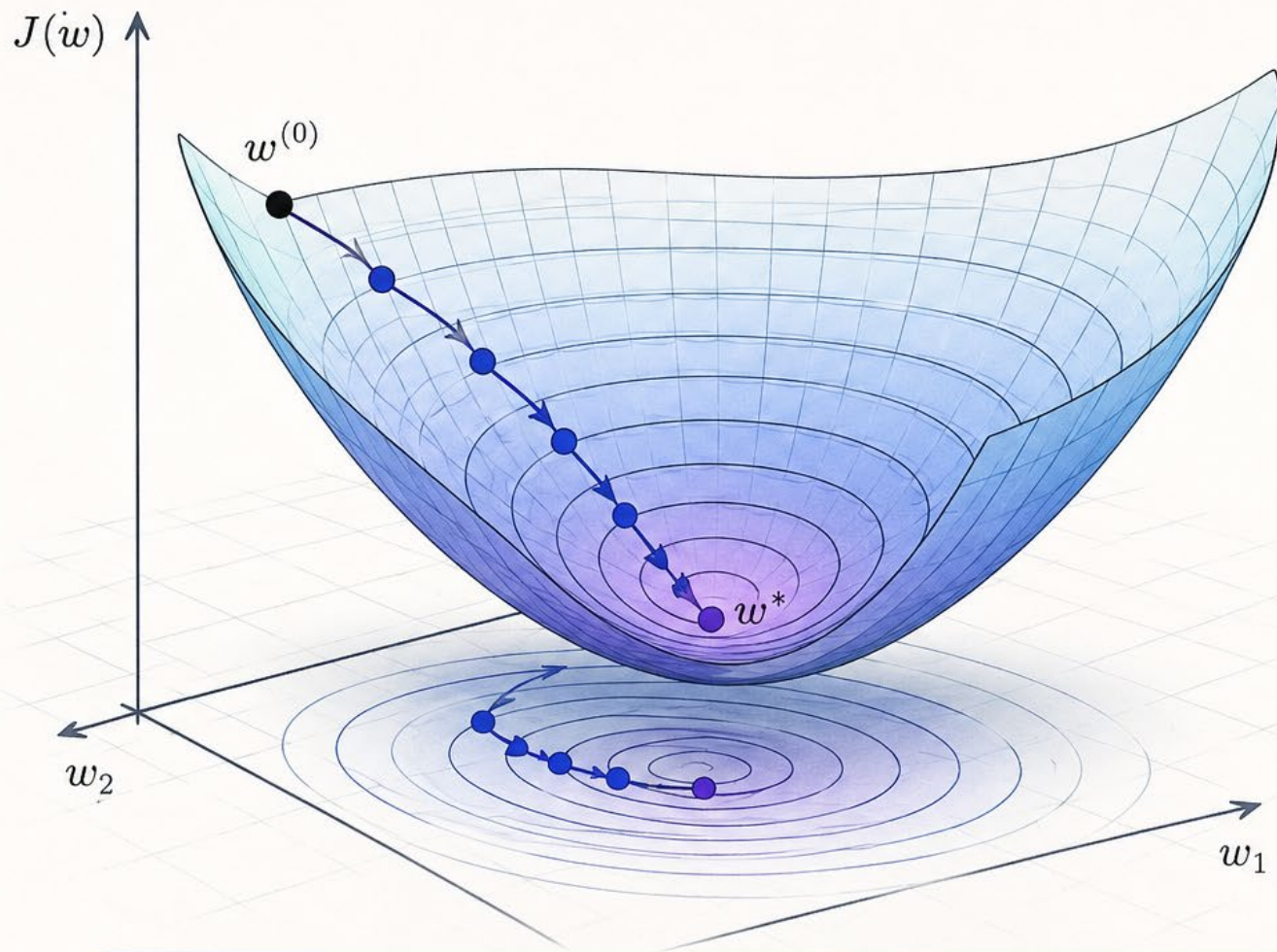


GROUP EXERCISE SOLUTIONS

1. What would be a good convergence condition (also known as a stopping criterion) for terminating GD?
 - If the norm of the gradient is small
 - If the decrease in the objective function or the parameter values is small
 - A fixed number of iterations
 2. What are some potential problems with setting $\alpha_t = \frac{1}{t}$?
 - The step size may decay too rapidly
 3. How could you choose α_t to counter these problems?
 - Could choose α_t adaptively; e.g., use momentum
 4. Does the initialization θ_0 affect the result?
 - If J is convex, we will eventually find the global min although, the initialization may affect the number of iterations required. If J is not convex, we may end up at a local min instead
-

LEARNING SCHEDULE

- A way to help “settle” stochastic gradient descent
 - Gradually reduce the learning rate according to a function called the “learning schedule”
 - Additional Points:
 - Common schedules include:
 - Step decay: Reduce learning rate by a factor after a fixed number of epochs.
 - Exponential decay: Multiply learning rate by a constant factor each epoch.
 - Polynomial decay: Decrease learning rate following a polynomial function.
 - Modern techniques:
 - Cosine Annealing: Learning rate follows a cosine curve, often combined with restarts.
 - One-cycle policy: Increase learning rate initially, then decrease sharply for better convergence.
 - Adaptive methods (Adam, RMSProp): Adjust learning rate per parameter automatically based on gradient history.
-



Update rule:

$$w^{(t+1)} = w^{(t)} - \eta \nabla J(w^{(t)})$$

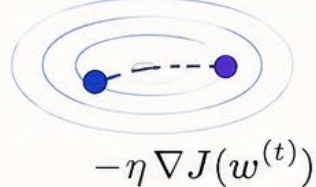
- $w^{(0)}$ Initial parameters
- $w^{(t)}$ Current parameters
- w^* Optimal parameters (minimum)
- ➔ Path of gradient descent

$$\nabla J(w) = \begin{bmatrix} \frac{\partial J}{\partial w_1} \\ \frac{\partial J}{\partial w_2} \end{bmatrix} \quad \text{Gradient (direction of steepest ascent)}$$

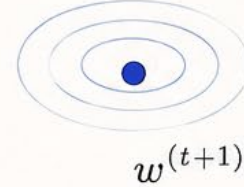
At each step t : 1. Compute gradient



2. Take a step



3. Arrive at new point



Repeat until convergence
(when updates become very small)



STOCHASTIC GRADIENT DESCENT (SGD)

- In many ML problems, we can write

$$J(\boldsymbol{\theta}) = \sum_{i=1}^n J_i(\boldsymbol{\theta})$$

- Thus

$$\nabla J(\boldsymbol{\theta}) = \sum_{i=1}^n \nabla J_i(\boldsymbol{\theta})$$

STOCHASTIC GRADIENT DESCENT (SGD)

- In GD, we calculate the gradient using all of the data
 - For large n , each step in the parameter space θ takes a lot of computations
 - Thus learning can occur slowly
- For SGD, we estimate the gradient using a small random sample of training inputs
 - Speeds up gradient descent and learning

STOCHASTIC GRADIENT DESCENT

- Pick a random set of $m < n$ training inputs
 - Referred to as a *mini-batch*
- Estimate the gradient using the samples from the mini-batch and take a step in the direction of the negative gradient
- If m is large enough, then the estimated gradient will be roughly equal to the gradient using the full data

STOCHASTIC GRADIENT DESCENT SUMMARY

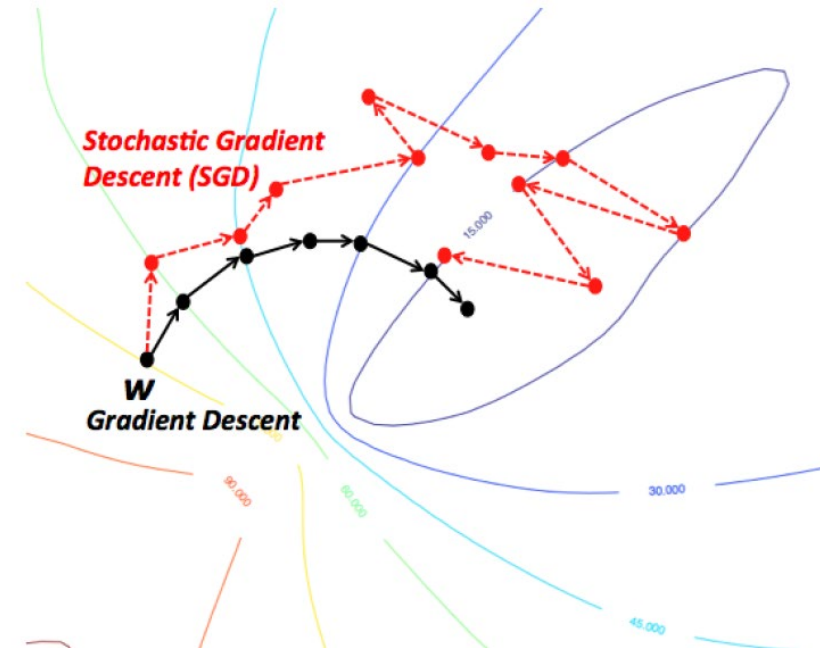
A training *epoch*:

1. Pick a random subset of the training data
 - Referred to as a mini-batch
2. Update the parameters using the gradient estimates from the mini-batch
3. Pick another random mini-batch from the remaining training points and repeat step 2
 - Repeat until all training inputs have been used

Repeat multiple epochs until stopping conditions are met

ANALOGY TO POLITICAL POLLING

- Much easier to carry out a poll than run a full election
- Similarly, it's much easier to estimate gradients from mini-batches than the entire training set
- Downside: gradient estimates will be noisier in SGD
- Generally ok as we only need to move in a general direction that decreases J
 - This can be an advantage when the problem is nonconvex
- In practice, SGD (or its variants) are used extensively in learning neural networks



GRADIENT DESCENT VARIATIONS

- Stochastic Gradient Descent (SGD)

- How it works:
 - Uses one random training instance at each step
 - Computes the gradient from that single instance
- **Pros:**
 - Fastest per update (very small computation)
 - Randomness helps escape local minima on complex cost surfaces
- **Cons:**
 - Updates are noisy; the algorithm never fully “settles”
 - True minimum is rarely reached—it oscillates around it

- Mini-Batch Gradient Descent

- How it works:
 - Uses a small random subset (mini-batch) at each step
 - Computes the gradient from that subset
 - **Pros:**
 - More stable than SGD (less noisy updates)
 - Faster than full batch GD on large datasets
 - Leverages vectorized operations for efficiency
 - **Cons:**
 - Still random, so convergence is not perfectly smooth
 - Requires tuning the batch size (too small = noisy, too large = slow)
-

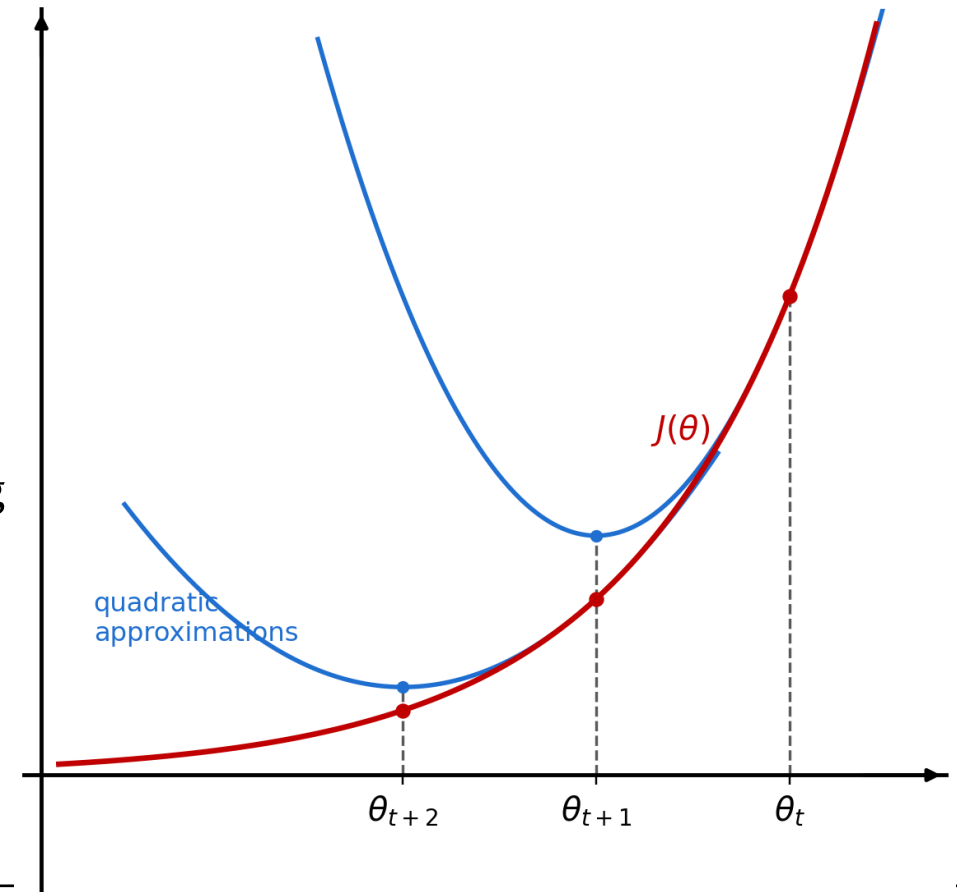
NEWTON'S METHOD

- Goal is to solve $\nabla J(\boldsymbol{\theta}) = \mathbf{0}$ for an objective function J
- Newton's method AKA Newton-Raphson:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - (\nabla^2 J(\boldsymbol{\theta}_t))^{-1} \nabla J(\boldsymbol{\theta}_t)$$

- Newton's method can be viewed as minimizing second order approximation:

$$J(\boldsymbol{\theta}) \approx J(\boldsymbol{\theta}_t) + \nabla J(\boldsymbol{\theta}_t)^T (\boldsymbol{\theta} - \boldsymbol{\theta}_t) + \frac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\theta}_t)^T \nabla^2 J(\boldsymbol{\theta}_t) (\boldsymbol{\theta} - \boldsymbol{\theta}_t)$$



NEWTON'S METHOD: INTUITION

- **GD fits a line; Newton fits a parabola**
 - Jumps straight to the minimum of its quadratic model of J
 - **The Hessian is an automatic learning rate**
 - $(\nabla^2 J)^{-1}$ shortens steps where curvature is high — no α to tune
 - **Fast when it works**
 - Exact in one step on a quadratic; quadratic convergence near a min
 - **But expensive and fragile**
 - $O(d^3)$ per step; Hessian may not be PD far from a min
 - Fixes: damping, trust regions, quasi-Newton (BFGS, L-BFGS)
-

GROUP EXERCISE

- One advantage of Newton's method is that it converges to a minimum in relatively few steps.
What is a disadvantage of Newton's method?
 - Answer: You have to invert a matrix (the Hessian) at each step
 - This can be very slow for high-dimensional data
 - Thus in practice, Newton's method is rarely used
 - However, Newton's method has provided inspiration for other optimization techniques that are faster to compute
-

GRADIENT DESCENT VARIATIONS

- Adam Optimizer
- How it works:
 - Maintains two moving averages:
 - First moment (m): average of gradients (like momentum).
 - Second moment (v): average of squared gradients
 - Applies bias correction to both moments.
 - Update rule:

$$\beta_{new} = \beta_{old} - \alpha \frac{\hat{m}}{\sqrt{\hat{v} + \epsilon}}$$